

PIA 402 Course Outline

PIA 402 – Project Identification and Demand Analysis

Where a project comes from before anyone has costed it.

School	04 – Public Investment
Hours	25 (indicative band 22-28)
Version	1.0 (2026-09-05)
Model	AICTEM PIAS, D:/EDIP/models/PIASModelling
Reference country	Utopia (UTP), UTD 2024 prices
Prerequisites	PIA 401. Identification is stage two of the cycle taught there, and this course assumes you can read a project record and a gate profile. No statistics beyond a growth rate and a ratio.
Audience	Planning units, sector policy analysts, project preparation teams, and anyone asked to justify why a project exists at all
Pass mark	60 %

What this course is

A project has to come from somewhere. In a well-run system it comes from a demonstrated need, expressed as a gap between what people have and what a service norm says they should have, located somewhere specific, and quantified as a demand that the project would meet. In a badly run one it arrives as a solution looking for a problem, and the needs assessment is written afterwards.

This course teaches the first half of the appraisal: everything that happens before a cost is attached. It covers the problem statement, the service norm and the gap, spatial need, the options analysis that the OPTIONS gate is blocking on, demand forecasting in the five forms PIAS implements, and the one quantity that a benefit is allowed to rest on – incremental demand.

Its central discipline is the counterfactual. A benefit is the difference between two futures, not a description of one, and the future without the project is not the present. The sample register makes this concrete: its without-project case grows at 3.20% a year in every one of the eleven projects, so crediting a project with the difference between today and year twenty would take credit for population growth that was going to happen anyway. In 10 of the eleven, the incremental demand is *negative* in the first year, because the asset is not yet delivering while the counterfactual has already grown.

The other discipline is that a need is spatial. A district-level average conceals the sub-district where nobody is served, and a project sited on an average is sited nowhere in particular.

What a graduate can do

Write a problem statement that a needs assessment could test; derive a service gap from a norm and a coverage figure; explain why a spatial need index is not a district average; state the five demand-forecast methods PIAS implements and choose between them on the evidence; compute incremental demand and say why it is the only quantity a benefit may rest on; identify a demand forecast supplied with no without-project case and state what that prevents; run an options analysis with a genuine do-minimum; classify a benefit by its valuation basis; and distinguish an output from an outcome without deriving one from the other.

What this course does not claim

This course does not value benefits. It computes incremental demand and stops there; the unit value that turns demand into money belongs to PIA 404, and the shadow prices behind that unit value belong to PIA 406.

It does not teach spatial analysis. It explains what a spatial need index is, why a district average is not one, and reads the outputs of the PIAS spatial layer, but the geometry, the accessibility calculations and the overlay operations are PIA 410.

It does not supply demand elasticities, growth rates or service norms for any sector. The Unit Cost and Service Norms database is a separate AICTEM system, and where this course needs a norm it says which system would supply it and what happens when that system does not answer.

It does not claim the sample register demonstrates good practice throughout. Its counterfactual is sound and is the course's best teaching material, but every one of its 244 demand rows uses a single method, **population**, and there are 0 recorded alternatives across eleven projects – which is consistent with the OPTIONS gate being PENDING on all of them. The course teaches the other four demand methods from the implemented equations rather than from a register example, and says so.

Learning outcomes

1. Write a problem statement that a needs assessment could test, and distinguish it from a description of a proposed solution.
2. Derive a service gap from a norm and a coverage figure, and say which system supplies the norm.
3. Explain why a spatial need index is not a district average, and what a district average conceals.
4. Read the spatial record of a project and say whether it is located at all.
5. State the five demand-forecast methods PIAS implements and choose between them on the evidence available.
6. Compute a demand forecast by each of the five methods and reconcile them.
7. Compute incremental demand and explain why it is the only quantity a benefit may rest on.
8. Identify a demand forecast supplied with no without-project case, and state exactly what that prevents.
9. Run an options analysis with a genuine do-minimum, and say why the do-minimum is not the do-nothing.
10. Classify a benefit by its valuation basis, and distinguish an output from an outcome without deriving one from the other.

Module structure

Module	Title	Sessions	Hours	Laboratory	Manual	Assessment
M01	Where a project comes from	1	1.25	–	1-2	Quiz M01
M02	The problem statement and the needs assessment	1	3.25	Lab 01	3	Quiz M02
M03	Service norms and the gap	1	1.50	–	4	Quiz M03
M04	Spatial need	1	3.50	Lab 02	5	Quiz M04
M05	Options analysis	1	1.50	–	6	Quiz M05
M06	Demand: the five methods	2	3.50	Lab 03	7-8	Quiz M06
M07	Incremental demand and the counter-factual	1	3.50	Lab 04	9	Quiz M07
M08	Benefit categories and valuation basis	1	1.50	–	10	Quiz M08
M09	Outputs, outcomes and what may not be derived	1	1.50	–	11	Quiz M09
M10	The identification note	1	3.50	Lab 05	12	Quiz M10

Laboratories

Lab	Title	Hours	Deliverable
1	Write a problem statement that could be false	2.0	A rewritten problem statement with its deletion test
2	Locate the gap, and build the short list that does not exist	2.0	A located gap and a screened short list

Lab	Title	Hours	Deliverable
3	Forecast the same demand five ways, and reconcile them	2.0	Five forecasts, reconciled, with a justified choice
4	The counterfactual, two ways, and what the difference costs	2.0	The counterfactual computed twice, with the error priced
5	Write the identification note, ending with what is missing	2.0	A complete identification note

Assessment

Component	Weight
Knowledge quizzes	15 %
Laboratory exercises	30 %
Applied assignments	25 %
Capstone / final project	30 %

Applied assignments

ID	Title	Weight	Deliverable
A1	The problem statement, rewritten	20	Marked on falsifiability and on the classification of the output
A2	The located gap and the short list	30	Marked on the costed do-minimum and the recorded screening reasons
A3	The counterfactual, priced	30	Marked on the arithmetic and on the treatment of the negative year
A4	The statement of absence	20	Marked on whether each absence names the gate or calculation it blocks

Final project

A complete identification note for one project: the problem statement, the service norm and the gap it implies, the spatial distribution of that gap, an options long list reduced to a short list with a genuine do-minimum, a demand forecast with an explicit without-project case and the method stated, the incremental demand that follows, and a statement of every quantity the note could not supply and what that prevents downstream.

Folder contents

Every course in the Academy carries the same section 5 folder. See `README.md` for what each item holds and `COURSE_BUILD_REPORT.md` for what was measured.